PUBLIC ABSTRACT · RELATIONAL DISTINGUISHABILITY / BASAL COGNITION

Relational Distinguishability in a Minimal Non-Neural Carrier

An intervention test through history-dependent repair

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THE BIOLOGICAL QUESTION AND PROPOSED POSTULATE

Work on anatomical homeostasis and basal cognition treats morphogenesis as collective problem-solving in morphospace: tissues can recover large-scale organization after perturbation, while physiological networks retain history not reducible to current anatomy alone [1–4]. **Can later repair depend on history retained in the relational organization of the same substrate that conducts repair?**

Relational Distinguishability (proposed postulate): “No relation is physically meaningful unless its sides are distinguishable.” Here, two histories count as physically different only if their difference changes later repair, survives relation-preserving address relabeling, and disappears when the relevant relational placement is shuffled. The link to relativity is programmatic, not a derivation: the tested invariance is storage-address relabeling, not a spacetime transformation.

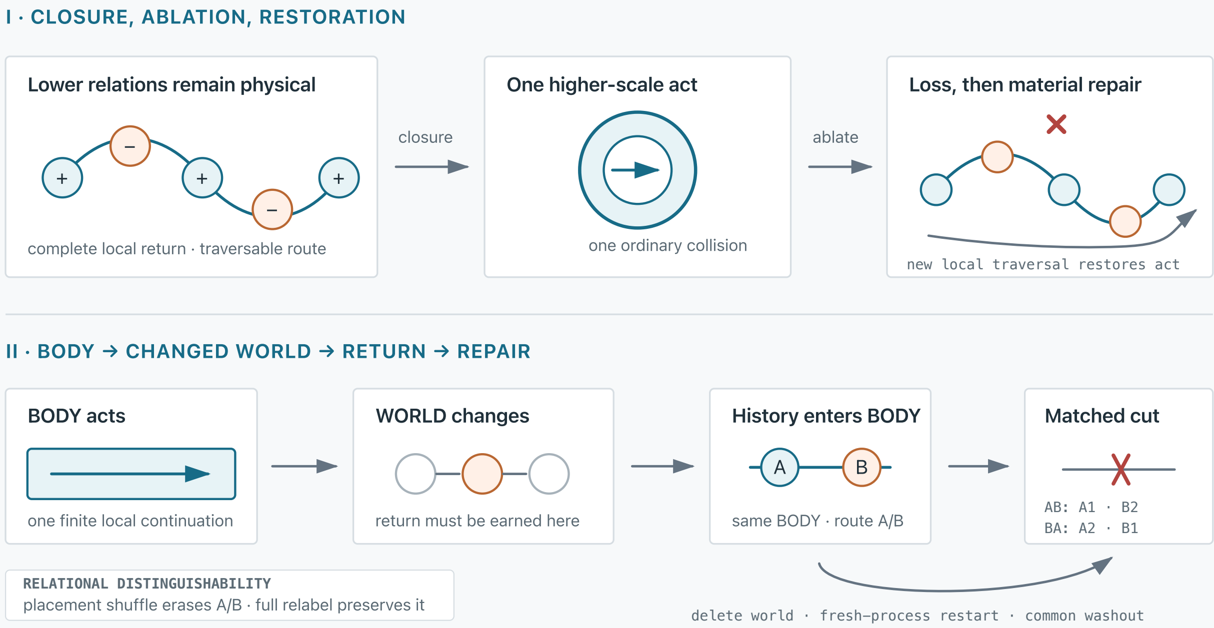

Figure 1. The carrier tests relational distinguishability in two intervention courts: an upper relation earned by prior closure must disappear under ablation and return through the surviving lower body; an earned body–world history must survive world deletion, fresh-process restart, matched washout, and renewed injury.

OPERATIONAL CRITERION

A relational history must remain distinguishable in future conduct under intervention—not merely classify a trace.

The harness fixes the A/B return opportunity before execution; it does not select a relation online.

1. BODY action changes a particular WORLD record.
2. Return can emerge only from that changed record.
3. The original world is deleted; BODY alone restarts.
4. Matched washout and the same new injury follow.
5. Relabel must preserve A/B; placement shuffle must erase it.

1 rule

collision kernel SHA-256
`0d32047eecb3…`

THE CARRIER

A deterministic body stores 1,024 writable records with explicit local relation handles. One rule updates one linked encounter at a time. The harness prepares topology, injury, worlds, and readout; A/B placement is fixed before the run. There is no global sweep, coordinate geometry, target register, reward, planner, or score-dependent update. The five-record world uses the same digital physics as BODY—an explicit limitation. Frozen audit: 21 tests · world-lineage source SHA-256: `2ac3304f0a95…`

History-dependent repair

A and B start from the same injured 513-record body. Equal-resource worlds differ only in where an earned final return enters two prepared routes. Both recover 1,024 records and the same current upper conductance. After world deletion, fresh-process restart, identical washout, and the same cut, A restores the upper entry in episode 1 of AB and 2 of BA; B gives the mirror pattern.

Interventions on relational distinguishability

Full address relabeling with relations preserved retains body organization and future A/B behavior. Shuffling all 257 candidate return-relation placements erases A/B while preserving their multiset, material extent, current upper conductance, and other relation classes. Separately, ablating the upper relation after closure changes one-collision PASS to HALT; one traversal of the surviving 12-place cycle rebuilds PASS.

A1 · B2
upper-entry birth episode in AB; mirrored in BA

4 / 4
prespecified deterministic initializations; not independent replicates

112
paired G1/G2 world-continuity controls

8 + 8
placement-shuffle + full-relabel interventions

What the result supports

A bounded constructive result: A/B histories remain relationally distinguishable after world deletion, restart, matched washout, and renewed injury. Return-relation placement is interventionally necessary for that distinction, while relation-preserving address relabeling preserves it. The interventions identify a distribution-dependent relation class, not a small memory locus.

What remains unresolved

Topology, closure affordance, routes, injury, and readout are engineered. The prepared organization may itself be a rudimentary distributed target or only a path-dependent attractor. This is not a biological model, independent replication, evidence of basal cognition, or evidence that morphogenesis lacks target states.

QUESTION FOR BIOLOGY

Would you regard the topology-encoded closure affordance and history-bearing relational state as a **minimal distributed pattern memory or homeostatic target, only a path-dependent attractor, or as both?** What single perturbation would best distinguish their physical roles?

REFERENCES

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